

S M Taslim Uddin Raju

📍 West Haven, CT, USA

✉️ smturaju@uwaterloo.ca

☎️ +1 (203) 410-1912

🌐 [raju32742.github.io](https://github.com/raju32742)

in [smturaju](#)

🔗 [raju32742](#)

Research Interests

My research interests lie at the intersection of **Computational Biology**, **Machine learning**, and **Deep learning** for healthcare. My research focuses on developing multimodal machine learning systems for healthcare by integrating Graph Neural Networks (GNNs), Vision Transformers (ViTs), and Large Language Models (LLMs). I specialize in digital pathology and medical imaging for cancer diagnosis and classification, leveraging deep learning on histopathological whole-slide images (WSIs) to enable automated tissue analysis and diagnostic report generation. I also work on non-invasive physiological monitoring through smartphone-based Photoplethysmography (PPG) for estimating vital biomarkers such as hemoglobin and glucose levels. My work emphasizes real-world applicability, bridging advanced AI techniques with practical healthcare needs.

Education

- 09/2023 – 04/2025 **University of Waterloo, Waterloo, Ontario, Canada**
MAsc. in Electrical and Computer Engineering (ECE) – Research | GPA: - 85%
Thesis: [Advanced AI for Histopathological Whole Slide Image Classification and Captioning](#)
Specialized: Pattern Analysis and Machine Intelligence (Now Artificial Intelligence)
Supervisor: Prof. Fakhri Karray
Core Courses: Introduction to Machine Learning, Advanced Deep Learning, Data and Knowledge Modelling and Analysis
- 06/2019 – 12/2022 **Khulna University of Engineering & Technology, Khulna, Bangladesh**
Master of Science in Computer Science and Engineering (CSE) – Research | GPA: - 4.00/4.00
Thesis: [Hemoglobin and Glucose Levels Estimation Techniques Using Optimal PPG Characteristic Features of Smartphone Videos](#)
Supervisor: Prof. M.M.A Hashem
Core Courses: Advance Deep Learning, Bioinformatics, Soft Computing, Advance Natural Language Processing
- 12/2014 – 02/2019 **Khulna University of Engineering & Technology, Khulna, Bangladesh**
Bachelor of Science in Computer Science and Engineering (CSE) | GPA: - 3.85/4.00 (3rd Position)
Thesis: [A Study on Non-Invasive Hemoglobin Measurement Techniques](#)
Supervisor: Prof. M.M.A Hashem
Core Courses: Machine Learning, Artificial Intelligence, Natural Language Processing, Biomedical Engineering, Data Structures, Operating Systems, Algorithms, Database, Data Mining, Computer Networks, Computer Security, Computer Graphics, Compiler, Software Engineering

Professional Experiences

- 01/2024 – 04/2025 **University of Waterloo, Waterloo, ON**
Machine Learning Graduate Research Assistant, Pattern Analysis and Machine Intelligence (Now AI) Lab
 - Advanced research in caption generation from histopathological whole slide image (WSI) through Transformer and UNet-Based Adversarial Autoencoder architectures.
 - Specialized in microscopic WSI analysis, developing Vision Transformers and Graph Neural Networks for advanced digital pathology tasks, such as classification and caption generation with LLMs.
- 02/2020 – 08/2023 **Khulna University of Engineering & Technology, Khulna, Bangladesh**
Lecturer, Department of Computer Science and Engineering (CSE)
 - Courses: CSE 1101 – Structure Programming, CSE 1102 – Structure Programming Laboratory, CSE 2113 – Computer Architecture and CSE 4112 – Machine Learning.
 - Taught undergraduate courses, supervised student projects, and mentored students through academic and technical guidance to support their learning and development.
- 03/2019 – 01/2020 **Eastern University, Dhaka, Bangladesh**
Lecturer, Department of Computer Science and Engineering
 - Courses: 06131205 – Structure Programming, 06131206 – Structure Programming Laboratory, 07142213 – Computer Architecture, 05413109 – Numerical Methods.
 - Delivered undergraduate lectures, designed course materials and assessments, and guided students through academic support.

Research Positions

- 09/2023 – 04/2025 **University of Waterloo, Waterloo, ON**
Machine Learning Graduate Research Assistant, Pattern Analysis and Machine Intelligence (Now AI) Lab
 - Advanced research in caption generation from histopathological whole slide image (WSI) through transformer and UNet-based Adversarial Autoencoder architectures.

	Specialized in microscopic WSI analysis, developing Vision Transformers and Graph Neural Networks for advanced digital pathology tasks, such as classification and caption generation with LLMs.
02/2020 – 12/2021	Khulna University of Engineering & Technology, Khulna, Bangladesh Graduate Researcher, Department of Computer Science and Engineering (CSE) - Pioneered a non-invasive 850 nm NIR-LED wearable device integrated with a smartphone to capture fingertip videos dataset and PPG signal extraction. - Engineered advanced ML pipelines – leveraging Deep Neural Networks (DNNs) and Multigene Genetic Programming (MGGP) – to estimate blood glucose, and hemoglobin levels from PPG extracted features.
03/2020 – 12/2020	Researcher, Artificial Intelligence in Medical Image Computing Lab - Collaborated on designing skip connections to regain the local information from shallower layers for heart segmentation in CT and MRI images. - Implemented residual skip connection consisting of a convolutional and regularizing path to compensate lost spatial information in the segmentation and localization.
02/2021 – 12/2021	OneBlood Organization, Florida, USA Collaborative Researcher, Biomedical Data Analysis Project - Analyzed donor behavior patterns and modeled successful donation outcomes using machine learning techniques. - Developed a forecasting pipeline to support decision-making in blood supply chain management.
03/2020 – 12/2020	King Saud University, Riyadh, Saudi Arabia Collaborative Researcher - Collaborated on developing ensemble-based machine learning frameworks to improve demand forecasting accuracy for industrial datasets. - Implemented preprocessing pipelines, feature selection, data normalization, and grid-search-based hyperparameter tuning for model optimization.

Major Projects

05/2024 – 04/2025	MMFformer: Multimodal Fusion Transformer Network for Depression Detection University of Waterloo, Pattern Analysis and Machine Intelligence Lab - Developed a transformer-based multimodal framework integrating video, audio, and textual cues for early depression detection from social media content - Achieved F1-score improvements of 13.92% on the D-Vlog dataset and 7.74% on the LMVD dataset, surpassing existing state-of-the-art models
09/2024 – 05/2025	GNN-ViTCap: Microscopic Whole Slide Image Classification and Captioning University of Waterloo, Pattern Analysis and Machine Intelligence Lab - Designed integrated framework combining graph neural networks and LLMs for whole slide images classification and captioning - Achieved high accuracy with BLEU-4 = 81.1% and METEOR = 56.7%, with BioMedGPT in image captioning
05/2024 – 08/2024	LLM-Q&A: Automated Medical Q & A Systems Using Fine-Tuned Large Language Models University of Waterloo, Pattern Analysis and Machine Intelligence Lab - Implemented an automated medical Q&A system by fine-tuning LLM models such as GPT-2, Llama2, Bloom, and T5 - Evaluated using BLEU and ROUGE metrics, with T5 showing superior performance in generating accurate medical answers
01/2020 – 12/2022	Non-Invasive Blood Component Levels Estimation Using Smartphone Fingertip Video Khulna University of Engineering & Technology - Introduced a non-invasive method for monitoring Glucose and Hemoglobin levels using Smartphone video and NIR LED device - Generated PPG signal, extracted the PPG features and fed the features to DNN-based models to estimate blood component levels
06/2019 – 08/2020	Dermo-DOCTOR Khulna University of Engineering & Technology, Artificial Intelligence in Medical Image Computing Lab - Engineered a CNN-based network with class rebalancing for state-of-the-art lesion detection on ISIC-2016 and ISIC-2017 datasets - Integrated segmented lesions and dual-encoder CNN features to enhance lesion detection across varied conditions

Publications

C = Conference, J = Journal Paper

Published/Accepted Manuscripts

- (J1) N. Hasan, S. B. Shuvo, M. H. Ankon, **S. T. U. Raju**, and N. Siddique, "TransfusionNet: Framework for Cervical Cancer Detection using Deep Learning with Multi-level Bi-Fusion and Aggregated-Fusion", *Results in Engineering*, Elsevier, p107174, vol. 28, Dec. 2025.
- (C1) S. T. Uddin, W. Mostary, M. N. Sakib, **S. T. U. Raju**, and A. R. Mohammad, "Enhancing Automated Medical Question-Answer Systems Using Fine-Tuned Large Language Models." In *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, pp. 1-6. IEEE, 2025.
- (C2) M. R. Haque, M. M. Islam, **S. T. U. Raju**, H. Altaheri, L. Nassar, and F. Karray, "MMFformer: Multimodal Fusion Transformer Network for Depression Detection", *International Conference on Systems, Man, and Cybernetics*, IEEE, Austria, 2025. [Accepted]
- (C3) M. R. Haque, M. M. Islam, **S. T. U. Raju**, H. Altaheri, L. Nassar, and F. Karray, "Multimodal Depression Detection through Mutual Transformer", *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, IEEE, Austria, 2025
- (C4) **S. T. U. Raju**, M. M. Islam, M. R. Haque, H. Altaheri, and F. Karray, "GNN-ViTCap: GNN-Enhanced Multiple Instance Learning with Vision Transformers for Whole Slide Image Classification and Captioning", *International Joint Conference on Neural Networks (IJCNN 2025)*, 30 June - 5 July 2025, IEEE, Rome, Italy [**Thesis Work**]
- (C5) **S. T. U. Raju**, A. R. Mohammad, M. M. Islam, and F. Karray, "TransUAAE-CapGen: Caption Generation from Histopathological Patches through Transformer and UNet-Based Adversarial Autoencoder", *IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC)*, IEEE, Borneo Convention Centre Kuching, Sarawak, Malaysia, 6 - 10 Oct., 2024 [**Thesis Work**]
- (J2) **S. T. U. Raju**, S. A. Dipto, M. I. Hossain, M. A. S. Chowdhury, F. Haque, A. T. Nashrah, A. Nishan, M. M. H. Khan, and M. Hashem, "DNN-BP: A Novel Framework for Cuffless Blood Pressure Measurement from Optimal PPG Features using Deep Learning Model," *Medical & Biological Engineering & Computing*, Springer, pp. 1–22, 2024.
- (J3) A. Nishan, **S. T. U. Raju**, M. I. Hossain, S. A. Dipto, S. T. Uddin, A. Sijan, M. A. S. Chowdhury, A. Ahmad, and M. M. H. Khan, "A Continuous Cuffless Blood Pressure Measurement from Optimal PPG Characteristic Features using Machine Learning Algorithms," *Heliyon*, Elsevier, vol. 10, no. 6, 2024.
- (C6) M. Jamil, S. Oman, **S. T. U. Raju**, and F. Soshi, "A Novel Framework for Enhancing Sensor Data Analysis: Label-Preserving Augmentation and Probabilistic Balancing", *26th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, 13-15 Dec. 2023.
- (C7) H. Neha, S. Sakib, F. Sadaf, and **S. T. U. Raju**, "Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way", *26th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, 13-15 Dec. 2023.
- (C8) L. Hossain, I. Hossain, **S. T. U. Raju**, MS Salim, J. Saha, "A Novel Technique for Classification of Motor Imagery EEG Signal Based on Deep Learning Approaches," *2nd International Conference on Big Data, IoT and Machine Learning (BIM 2023)*, Springer, Dhaka, Bangladesh, 6-8 Sep. 2023
- (C9) S. Oman, M. Jamil, and **S. T. U. Raju**, "BCL: A Branched CNN-LSTM Architecture for Human Activity Recognition Using Smartphone Sensors," *International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM 2023)*, IEEE, DUET Gazipur, Bangladesh, 16-17 Jun., 2023.
- (C10) R. T. H. Promi, R. A. Nazri, M. S. Salim, and **S. T. U. Raju**, "A Deep Learning Approach for Non-Invasive Hypertension Classification from PPG Signal" *International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM 2023)*, IEEE, DUET Gazipur, Bangladesh, 16-17 Jun. 2023.
- (C11) **S. M. T. Raju**, and M. Hashem, "Real-Time Hemoglobin Measurement Using Smartphone Video and Artificial Neural Network," *International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2022)*, IEEE, RUET, Rajshahi, Bangladesh, 29 – 31 Dec. 2022.
- (C12) **S. M. T. Raju**, and M. Hashem, "DNN Based Blood Glucose Level Estimation Using PPG Characteristic Features of Smartphone Videos," *25th International Conference on Computer and Information Technology (ICCIT 2022)*, IEEE, Cox's Bazar, Bangladesh, 17-19 Dec. 2022.
- (C13) T. Ahamed, M. N. Islam, **S. T. U. Raju**, and M. Hashem, "Blood Donor Arrival Forecasting Using Regression Model and Analysis of Donor Behavioural Pattern," *25th International Conference on Computer and Information Technology (ICCIT 2022)*, IEEE, Cox's Bazar, Bangladesh, 17-19 Dec. 2022.
- (J4) **S. T. U. Raju**, A. Sarker, A. Das, M. M. Islam, M. S. Al-Rakhani, A. M. Al-Amri, T. Mohiuddin, and F. R. Albogamy, "An Approach for Demand Forecasting in Steel Industries Using Ensemble Learning," *Complexity*, vol. 2022, no. 1, p. 9928836, 2022.
- (C14) F. Sadaf, **S. T. U. Raju**, and A. Muntakim, "Offline Bangla Handwritten Text Recognition: A Comprehensive Study of Various Deep Learning Approaches" *2021 3rd International Conference on Electrical & Electronic Engineering (ICEEE 2021)*, IEEE, RUET, Rajshahi, Bangladesh, 22 – 24 Dec., 2021.
- (C15) A. Ghosh, A. A. Hossain, and **S. T. U. Raju**, "Classification of Diabetic Retinopathy Using Few-Shot Transfer Learning from Imbalanced Data" *2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS)*, IEEE, Tamilnadu, India, 19 – 20 Mar., 2021.

- (J5) M. K. Hasan, S. Roy, C. Mondal, M. A. Alam, M. T. E. Elahi, A. Dutta, **S. T. U. Raju**, M. T. Jawad, and M. Ahmad, "Dermo-Doctor: A Framework for Concurrent Skin Lesion Detection and Recognition Using a Deep Convolutional Neural Network With End-to-End Dual Encoders", *Biomedical Signal Processing and Control*, Elsevier, vol. 68, p. 102661, 2021.
- (J6) M. R. Haque, **S. T. U. Raju**, M. A.-U. Golap, and M. Hashem, "A Novel Technique for Non-Invasive Measurement of Human Blood Component Levels from Fingertip Video Using DNN Based Models" *IEEE Access*, vol. 9, pp. 19 025–19 042, 2021.
- (J7) M. A. Golap, **S. T. U. Raju**, M. R. Haque, and M. Hashem, "Hemoglobin and Glucose Level Estimation From PPG Characteristic Features of Fingertip Video Using MGGP-Based Model," *Biomedical Signal Processing and Control*, Elsevier vol. 67, 2021.
- (J8) S. M. A. Ullah, M. M. Islam, S. Mahmud, S. Nooruddin, **S. T. U. Raju**, and M. R. Haque, "Scalable Telehealth Services to Combat Novel Coronavirus (COVID-19) Pandemic" *SN Computer Science*, Springer, vol. 2, no. 1, pp. 18, Jan. 2021.
- (J9) M. M. Islam, S. M. A. Ullah, S. Mahmud, and **S. T. U. Raju**, "Breathing Aid Devices to Support Novel Coronavirus (COVID-19) Infected Patients," *SN Computer Science*, Springer, vol. 1, no. 5, pp. 274, Aug. 2020.
- (C16) **S. T. U. Raju**, and M. S. Rahman, "Horizontal Vertical and SuperQueen Parity (HVSQ) Method for Soft Error Tolerance," *2020 IEEE Region 10 Symposium (TENSYP)*, IEEE, Dhaka, Bangladesh, pp. 1734-1737, 5-7 Jun., 2020.

Posters/Workshop Publications

- (P1) M. M. Islam, M. R. Haque, **S. T. U. Raju**, and F. Karray, "FusionEnsemble-Net: An Attention-Based Ensemble of Spatiotemporal Networks for Multimodal Sign Language Recognition", *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Workshop, Honolulu, Hawaii, USA, Oct. 20th, 2025
- (P2) M. R. Haque, M. M. Islam, **S. T. U. Raju**, and F. Karray, "A Signer-Invariant Conformer and Multi-Scale Fusion Transformer for Continuous Sign Language Recognition", *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Workshop, Honolulu, Hawaii, USA, Oct. 20th, 2025
- (P3) **S. T. U. Raju**, and M. M. A. Hashem, "Development of a Novel Non-invasive Smartphone-Based Blood Components Estimation Technique Using Python," PyCon US 2023, Salt Lake City, USA, 19-27 April 2023

Book Chapter

- (B1) M. M. Islam, **S. T. U. Raju**, S. Nooruddin, F. Karray, and G. Muhammad, "Internet of Health Things: an introduction." *In Blockchain and Digital Twin for Smart Healthcare*, pp. 19-44. Elsevier, 2025.

Submitted Manuscripts

- (S1) **S. T. U. Raju**, M. R. Haque, M. M. Islam, H. Altaheri, and F. Karray, "ClusGNN-ViT: Integrating Deep Embedded Clustering and GNN-MIL for Microscopic Image Captioning and Diagnosis," *IEEE Transactions on Medical Imaging*, IEEE [Manuscript Number: TMI-2025-3074]
- (S2) **S. T. U. Raju**, A. Das, A. Sarker, S. T. Uddin, M. I. Hossain, and M. P. Mollah, "EwvEn: An Enhancing Weighted Voting Ensemble Algorithm for Demand Forecasting of Steel Industry," *IEEE Transactions on Industrial Informatics*, IEEE [Manuscript Number: TII-25-4967]

Leadership and Mentorship

2024	Mentor, Graduate Mentorship Program, University of Waterloo
	Guided incoming graduate students on research planning, academic adjustment, and career development. [PDF]
2021 – 2023	Capstone Project Supervisor, Khulna University of Engineering & Technology
	- Supervised an undergraduate 4th-year capstone team on deep learning for biomedical data analysis; the work led to a publication. [PDF] (C10) - Guided an undergraduate capstone team on EEG-based deep learning research that led to a publication. [PDF] (C8)
2022	Research Mentor, Khulna University of Engineering & Technology
	Supervised 3rd year undergraduates on deep learning and sensor-based projects, fostering research skills and teamwork. [PDF] (C6) (C9)
2021	Mentor, System Development Project, Khulna University of Engineering & Technology
	Supervised 3rd-year students in developing smartphone-based solutions. [PDF]
2017 – 2018	Student Coordinator, Department of CSE, Khulna University of Engineering & Technology
	Coordinated campus tours, student volunteers, and departmental outreach.

Services and Outreach

Journal Reviews

- IEEE Transactions on Industrial Informatics, 2025
- Journal of Biomedical and Health Informatics, 2025
- IEEE Access, 2022, 2025
- IEEE Sensors Journal, 2025
- Discover Computing – Springer Nature, 2025
- International Journal of Intelligent Systems – Wiley, 2025
- Technology and Health Care – Sage Journals, 2025

- Computer Methods in Biomechanics and Biomedical Engineering, 2024 – 2025
- Biomedical Engineering: Applications, Basis and Communications, 2025
- Journal of International Medical Research, 2024
- Scientific Reports – Nature, 2024
- Journal of Engineering – Wiley, 2024
- Signal, Image and Video Processing – Springer Nature, 2024
- IEEE Sensors Letters, 2022
- Plos One 2022

Conferences Reviews

- IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC) 2024 – 2025
- International Joint Conference on Neural Networks (IJCNN 2025)
- 7th International Conference on Electrical Information and Communication Technology (EICT), 2025

Volunteering

- Volunteer Member to assisting with conference logistics, attendee support for 5th International Conference on Electrical Information and Communication Technology (EICT) [2021]
- **Instructor** for introductory workshop on C Programming in SGPIC (Special Group Interested in Programming Contest) [2016]
- Student Motivator and Examiner in **NHSPC** (National High School Programming Contest) [2016- 2017]

Fellowships, Honors, & Awards

2025	Best Poster Award , 1 st Multimodal Sign Language Recognition Workshop IEEE/CVF ICCV 2025
2025	• Recognized for contributions to multimodal sign language recognition as part of the CPAMI Team, UW [PDF]
2025	ICCV 2025, 2nd Place Award at Multimodal Sign Language Recognition Challenge for Unseen Sentences Task
2023 – 2025	• Recognized for high-performing sign language recognition among CPAMI Team, UW [Awarded \$750 USD] [PDF]
2023 – 2025	Graduate Research Students (GRS), University of Waterloo
	• Awarded for research in digital pathology and AI-driven whole-slide image analysis, covering full tuition, health insurance, and a research stipend [Funding: \$55000 CAD] [PDF]
2024	Scholar Award , IEEE International Conference on Systems, Man and Cybernetics (IEEE SMC) [Funding: \$500 USD]
2023	Scholar Award , PyCon US Poster Presentation, Salt Lake City, Utah USA [Funding: \$2000 (USD)] [PDF]
2023	Vice Chancellor Awards for top researcher, Khulna University of Engineering & Technology [Outstanding research contributions in Engineering faculty] [Funding: \$500 (USD)] [PDF]
2022	Travel Grant, 25th International Conference on Computer and Information Technology (ICCIT 2022), IEEE, Cox's Bazar, Bangladesh [Funding: \$400 USD]
2021	Travel Grant, 3rd International Conference on Electrical & Electronic Engineering (ICEEE 2021), IEEE, RUET, Rajshahi, Bangladesh [Funding: \$300 USD]
2017 – 2018	Dean's List , outstanding academic performance, KUET [CGPA: 3.93/4.00] [PDF]
2016 – 2017	Dean's List , outstanding academic performance, KUET [CGPA: 3.94/4.00] [PDF]
2015 – 2016	Dean's List , outstanding academic performance, KUET [CGPA: 3.90/4.00] [PDF]
2016 – 2019	Technical Scholarship, Khulna University of Engineering & Technology [Funding: \$250 (USD)/Year]

Skill & Tools

- **ML & DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn
- **Programming Languages:** Python (Primary), C, C++, C#, Java, SQL/MySQL, MATLAB
- **Development Tools & IDEs:** Jupyter Notebook, PyCharm, VS Code, Microsoft Visual Studio, GitHub, LaTeX/Overleaf, Figma
- **Soft Skills:** Teamwork, Interdisciplinary Collaboration, Cross-culture Collaboration, Analytical Problem-solving, Project Management, Presentation, Communication

References

Prof. Fakhri Karray

Department of Electrical and Computer Engineering
 Centre for Pattern Analysis and Machine Intelligence
 University of Waterloo
 200 University Avenue West, Waterloo, ON, Canada
 Email: karray@uwaterloo.ca
 Website: uwaterloo.ca/electrical-computer-engineering/profile/karray

Prof. M. M. A. Hashem

Department of Computer Science and Engineering
 Khulna University of Engineering & Technology
 Khulna-9203, Bangladesh.
 Email: hashem@cse.kuet.ac.bd
 Mobile: +8801714003949
 Website: https://www.kuet.ac.bd/cse/hashem/